SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE CODE: MLA0303
COURSE NAME: Reinforcement learning for Environment and Agent

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COURSE OUTCOME COVERED

CO3: *Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3, BL6).*

Assessment Tool Weightage: 25%

Dr DUMPALA PRASANTH

QUESTIONS: 10 **Total Marks: 50**

Annexure A

Data Interpretation **Duration: 1hr**

1. A Reinforcement Learning agent is trained for multiple episodes and the reward values increase gradually from 10 to 45. Interpret the trend in rewards and defend what indicates about the learning performance and policy improvement of the agent.
(5 Marks)

2. An epsilon-greedy agent is tested with different epsilon values such as 0.1, 0.3 and 0.5. The cumulative reward obtained is highest when epsilon is 0.1 and lowest when epsilon is 0.5. Justify and apply the concept of exploration vs exploitation to interpret why the reward changes with epsilon and identify which epsilon value provides better performance. **(5 Marks)**

3. Two Reinforcement Learning algorithms, Q-Learning and SARSA, are compared based on average rewards obtained over 100, 200 and 300 episodes. Q-Learning consistently shows slightly higher reward values than SARSA. Identify the performance difference between these algorithms and determine which algorithm converges faster based on the observed results. **(5 Marks)**

4. An RL agent is trained using different discount factor values such as 0.2, 0.5 and 0.9. The average reward increases as the discount factor increases. Discuss how the discount factor affects long-term reward optimization and justify which discount factor is most suitable for achieving better learning performance. **(5 Marks)**

5. The success rate of an RL agent increases from 45 percent at 50 episodes to 92 percent at 200 episodes. Discuss the improvement in success rate and interpret whether the agent has learned an optimal policy. Provide reasoning based on the trend of the data. **(5 Marks)**

6. An RL agent is trained in the CartPole environment with different learning rates (0.01, 0.05, 0.1). The agent achieves higher stability at 0.05 compared to the other values. Analyze how learning rate influences convergence and justify which value provides the best trade-off between speed and stability. **(5 Marks)**

7. A Q-learning agent is tested with different exploration strategies: ε-greedy, softmax, and random exploration. The cumulative reward is highest with ε-greedy. Compare the strategies and defend why ε-greedy achieves better performance in this scenario. **(5 Marks)**

8. An RL agent trained in a maze environment initially takes 50 steps to reach the goal, but after 300 episodes it reduces to 15 steps. Interpret the reduction in steps and evaluate whether this indicates convergence to an optimal policy. **(5 Marks)**

9. A SARSA agent is trained with discount factors of 0.4, 0.7, and 0.95. The agent achieves the highest long-term reward with 0.95. Discuss the role of discount factor in balancing immediate vs. future rewards and justify why 0.95 is most effective. **(5 Marks)**

10. An RL agent trained in a stochastic environment shows fluctuating rewards in early episodes but stabilizes after 500 episodes. Examine the reward trend and defend how stability indicates policy improvement and robustness of learning. **(5 Marks)**

Code of Conduct:

I _P.Sashank_(192425394_)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:P Sashank

RUBRICS FOR CALCULATION

Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to

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1. A Reinforcement Learning agent is trained for multiple episodes and the reward values increase gradually from 10 to 45. Interpret the trend in rewards and defend what indicates about the learning performance and policy improvement of the agent. *(5 Marks)*

Answer:

In Reinforcement Learning (RL), the reward received by the agent is one of the most important indicators of learning performance. The given data shows that the reward gradually increases from **10 to 45** over multiple training episodes. This increasing trend indicates that the agent is successfully learning from its interaction with the environment and continuously improving its decision-making ability.

During the initial episodes, the agent has little or no knowledge of the environment. It performs random actions through exploration, which often leads to lower rewards. As the number of episodes increases, the agent gains experience and updates its policy based on the rewards received. This learning process enables the agent to identify which actions lead to better outcomes and avoid actions that produce lower rewards.

The steady increase in rewards also indicates that the value function or Q-values are becoming more accurate. The agent is gradually selecting actions that maximize the expected cumulative reward instead of making random decisions. This shows that the policy is improving with every episode.

Another important observation is that there is no sudden decrease in reward values. A gradual increase demonstrates stable learning and indicates that the algorithm is effectively converging toward an optimal policy rather than oscillating between good and bad decisions.

Therefore, the increase in rewards from **10 to 45** clearly shows that the Reinforcement Learning agent has improved its learning performance, gained better knowledge of the environment, and developed a more efficient policy for achieving higher rewards.

2. An epsilon-greedy agent is tested with different epsilon values such as 0.1, 0.3 and 0.5. The cumulative reward obtained is highest when epsilon is 0.1 and lowest when epsilon is 0.5. Justify and apply the concept of exploration vs exploitation to interpret why the reward changes with epsilon and identify which epsilon value provides better performance. *(5 Marks)*

Answer:

The epsilon-greedy strategy is widely used in Reinforcement Learning to balance **exploration** and **exploitation**. Exploration means trying new actions to gain more knowledge about the environment, while exploitation means selecting the best-known action based on previous experience.

When the epsilon value is **0.1**, the agent explores only 10% of the time and exploits its learned knowledge for the remaining 90%. Since the agent mostly chooses the best actions while still performing limited exploration, it achieves the highest cumulative reward.

When epsilon is increased to **0.3**, the agent explores more frequently. Although exploration helps discover new strategies, excessive exploration reduces the number of optimal actions selected, leading to a moderate decrease in cumulative reward.

At **epsilon = 0.5**, the agent spends half of its time exploring random actions. Because many of these actions are not optimal, the cumulative reward becomes the lowest among the three values.

This experiment demonstrates the importance of maintaining a proper balance between exploration and exploitation. Too little exploration may prevent the discovery of better policies, while too much exploration reduces overall performance because the agent performs many unnecessary actions.

Among the given values, **epsilon = 0.1** provides the best balance. It allows the agent to exploit its learned knowledge efficiently while still exploring occasionally to avoid getting trapped in local optima. Therefore, **ε = 0.1 delivers the best overall learning performance and the highest cumulative reward.**

3. Two Reinforcement Learning algorithms, Q-Learning and SARSA, are compared based on average rewards obtained over 100, 200 and 300 episodes. Q-Learning consistently shows slightly higher reward values than SARSA. Identify the performance difference between these algorithms and determine which algorithm converges faster based on the observed results. *(5 Marks)*

Answer:

Q-Learning and SARSA are two popular Reinforcement Learning algorithms used for solving sequential decision-making problems. Although both algorithms learn action-value functions, they differ in the way they update their Q-values.

Q-Learning is an **off-policy** learning algorithm. It updates its Q-values using the maximum possible future reward regardless of the action actually selected by the current policy. Because of this optimistic update strategy, Q-Learning generally learns the optimal policy faster and achieves higher average rewards.

SARSA, on the other hand, is an **on-policy** learning algorithm. It updates its Q-values based on the actual action chosen by the agent. Since the update depends on the current policy, SARSA is more conservative and learns safer policies, especially in uncertain environments.

The observed results show that Q-Learning consistently achieves higher rewards after **100, 200, and 300 episodes**. This indicates that Q-Learning is improving its policy more rapidly and making better decisions than SARSA.

Higher average rewards also suggest that Q-Learning converges faster toward the optimal policy. SARSA may require more training episodes to achieve similar performance because it continuously considers the exploratory actions taken during learning.

Therefore, based on the given data, **Q-Learning demonstrates better performance, faster convergence, and more efficient policy optimization than SARSA.**

4. An RL agent is trained using different discount factor values such as 0.2, 0.5 and 0.9. The average reward increases as the discount factor increases. Discuss how the discount factor affects long-term reward optimization and justify which discount factor is most suitable for achieving better learning performance. *(5 Marks)*

Answer:

The **discount factor (γ)** determines how much importance an RL agent gives to future rewards compared to immediate rewards. It is one of the most important hyperparameters in Reinforcement Learning.

When the discount factor is **0.2**, the agent mainly focuses on immediate rewards. Although this allows the agent to achieve quick short-term gains, it often ignores future benefits, resulting in lower average rewards.

With a discount factor of **0.5**, the agent gives moderate importance to future rewards. It balances immediate and long-term rewards, leading to improved performance compared to $\gamma = 0.2$.

When the discount factor increases to **0.9**, the agent strongly considers future rewards while making decisions. Instead of choosing actions that provide immediate benefits, it learns strategies that maximize cumulative rewards over a longer period. As a result, the average reward becomes significantly higher.

A higher discount factor encourages better planning and allows the agent to identify sequences of actions that produce greater long-term success. This is particularly beneficial in complex environments where future consequences are important.

Therefore, **$\gamma = 0.9$** is the most suitable discount factor because it enables better long-term reward optimization, improves policy quality, and produces superior learning performance.

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5. The success rate of an RL agent increases from 45% at 50 episodes to 92% at 200 episodes. Discuss the improvement in success rate and interpret whether the agent has learned an optimal policy. Provide reasoning based on the trend of the data. *(5 Marks)*

Answer:

The success rate measures how frequently the Reinforcement Learning agent successfully completes its assigned task. In the given data, the success rate improves from **45% after 50 episodes** to **92% after 200 episodes**, showing substantial learning progress.
Initially, the success rate is relatively low because the agent has limited experience with the environment. During this stage, it performs many exploratory actions and frequently makes incorrect decisions.
As training continues, the agent gathers more experience, updates its value estimates, and improves its policy. The increasing success rate indicates that the agent is making better decisions and selecting optimal actions more consistently.
A success rate of **92%** means the agent completes its task successfully in almost every episode. This suggests that the policy has become highly effective and reliable.
Although a 92% success rate does not necessarily guarantee a perfectly optimal policy, it strongly indicates that the agent has nearly converged to the optimal solution. Additional training may produce only minor improvements because the learning curve has already reached a high level of performance.
Therefore, the observed trend clearly demonstrates continuous learning, policy improvement, and near-optimal decision-making ability.

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6. An RL agent is trained in the CartPole environment with different learning rates (0.01, 0.05, 0.1). The agent achieves higher stability at 0.05 compared to the other values. Analyze how learning rate influences convergence and justify which value provides the best trade-off between speed and stability. *(5 Marks)*

Answer:

The learning rate determines how much the Reinforcement Learning agent updates its knowledge after each interaction with the environment.
A learning rate of **0.01** causes very small updates to the Q-values or policy parameters. Although this leads to stable learning, convergence becomes very slow because the agent requires many more episodes to learn an effective policy.
A learning rate of **0.1** causes large updates. While learning becomes faster initially, the updates may become unstable, causing oscillations and inconsistent performance. Large learning rates may also prevent convergence if the agent continually overshoots the optimal values.
The learning rate of **0.05** provides the best compromise. It allows the agent to learn quickly while maintaining stable updates throughout training. The policy improves smoothly without experiencing significant fluctuations.

In the CartPole environment, maintaining balance requires precise decision-making. Stable parameter updates are essential for successful learning.
Therefore, **0.05** offers the best balance between convergence speed and stability, making it the most effective learning rate among the given values.

7. A Q-learning agent is tested with different exploration strategies: ε-greedy, softmax, and random exploration. The cumulative reward is highest with ε-greedy. Compare the strategies and defend why ε-greedy achieves better performance in this scenario. *(5 Marks)*

Answer:

Exploration strategies play an important role in Reinforcement Learning because they determine how an agent balances **exploration** (trying new actions) and **exploitation** (choosing the best-known action). Selecting an appropriate exploration strategy directly affects the cumulative reward and overall learning performance of the agent.
In this scenario, three exploration strategies are compared: **ε-greedy, softmax, and random exploration**. The results show that the **ε-greedy strategy achieves the highest cumulative reward**. This indicates that ε-greedy provides the most effective balance between exploring the environment and exploiting the learned policy.
The **ε-greedy strategy** selects the best-known action with a high probability while occasionally choosing a random action with probability ε. This allows the agent to continue discovering new actions without sacrificing overall performance. Because the agent mostly selects actions with high expected rewards, it accumulates greater rewards during training.
The **softmax strategy** assigns probabilities to actions based on their estimated values. Better actions are chosen more frequently, but lower-valued actions may still be selected occasionally. Although this strategy provides smoother exploration than random selection, it may spend more time exploring less rewarding actions, resulting in slightly lower cumulative rewards compared to ε-greedy.
In **random exploration**, every action has an equal chance of being selected regardless of its reward value. While this strategy helps the agent explore the entire environment, it often chooses poor actions even after learning has progressed. Consequently, the cumulative reward remains lower because the agent does not effectively utilize its learned knowledge.
The higher cumulative reward achieved by ε-greedy demonstrates that the agent successfully balances exploration and exploitation. It explores enough to discover better actions while exploiting its existing knowledge to maximize rewards. This

balance enables faster learning, improved policy development, and better convergence toward the optimal solution.
Therefore, among the three exploration strategies, **ε-greedy provides the best learning performance** because it achieves the highest cumulative reward, improves decision-making efficiency, and allows the agent to learn an optimal policy more effectively.

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8. An RL agent trained in a maze environment initially takes 50 steps to reach the goal, but after 300 episodes it reduces to 15 steps. Interpret the reduction in steps and evaluate whether this indicates convergence to an optimal policy.
(5 Marks)

Answer:

The number of steps taken by a Reinforcement Learning agent to reach its goal is an important performance metric. Fewer steps indicate that the agent has learned a more efficient path and is making better decisions while interacting with the environment.
Initially, the agent requires **50 steps** to reach the goal in the maze. This large number of steps suggests that the agent has very little knowledge of the environment and spends considerable time exploring different paths. During this stage, the agent often selects unnecessary or incorrect actions, resulting in longer paths and lower efficiency.
As training continues over **300 episodes**, the agent gains experience from repeated interactions with the maze. It learns which paths lead to the goal quickly and which paths should be avoided. The value function and policy are continuously updated, allowing the agent to make better decisions based on previous experience.
After sufficient training, the number of steps decreases significantly to **15**. This reduction indicates that the agent has successfully identified a shorter and more optimal path to the destination. The improved policy minimizes unnecessary movements, reduces the time required to reach the goal, and increases overall efficiency.
The consistent reduction in steps also suggests that the learning algorithm is converging toward an optimal policy. Convergence occurs when the agent's behavior becomes stable and further training produces only minor improvements. Since the agent now consistently reaches the goal using only 15 steps, it demonstrates that learning has become more effective and reliable.
The reduction in steps also improves computational efficiency and resource utilization because fewer actions are required to complete the task. This is particularly important in real-world applications such as robotic navigation,

autonomous vehicles, and warehouse automation, where efficient path planning reduces energy consumption and operating time.

Therefore, the decrease from **50 steps to 15 steps** clearly indicates successful learning, improved policy optimization, and convergence toward an optimal solution. The agent has learned to navigate the maze efficiently by selecting the shortest and most rewarding path.

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9. A SARSA agent is trained with discount factors of 0.4, 0.7, and 0.95. The agent achieves the highest long-term reward with 0.95. Discuss the role of discount factor in balancing immediate vs. future rewards and justify why 0.95 is most effective. *(5 Marks)*

Answer:

The **discount factor (γ)** is an important parameter in Reinforcement Learning that determines how much importance the agent gives to future rewards compared to immediate rewards. It helps the agent decide whether to focus on short-term benefits or long-term success while learning a policy.

In this scenario, the SARSA agent is trained using discount factors of **0.4, 0.7, and 0.95**, and the highest long-term reward is obtained when **γ = 0.95**. This observation indicates that considering future rewards leads to better overall performance.

When the discount factor is **0.4**, the agent mainly focuses on immediate rewards. Although it can quickly obtain short-term gains, it often ignores actions that may provide greater rewards in the future. As a result, the cumulative reward remains relatively low.

With a discount factor of **0.7**, the agent begins balancing immediate and future rewards. It considers the long-term consequences of its actions, leading to improved decision-making and higher cumulative rewards than $\gamma = 0.4$.

When the discount factor increases to **0.95**, the agent places significant importance on future rewards. This encourages it to learn strategies that maximize cumulative rewards over many future steps instead of focusing only on immediate outcomes. Consequently, the agent develops a more effective policy and achieves the highest long-term reward.

Since SARSA is an **on-policy learning algorithm**, it updates its values based on the actual actions taken by the current policy. A higher discount factor enables SARSA to evaluate future consequences more effectively, resulting in safer and more consistent learning.

Therefore, **γ = 0.95** is the most effective discount factor because it provides excellent long-term reward optimization, improves policy quality, and enables the agent to achieve superior learning performance while maintaining stable convergence.

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10. An RL agent trained in a stochastic environment shows fluctuating rewards in early episodes but stabilizes after 500 episodes. Examine the reward trend and defend how stability indicates policy improvement and robustness of learning. *(5 Marks)*

Answer:

In Reinforcement Learning, stochastic environments contain uncertainty, meaning that the same action may produce different outcomes at different times. Because of this randomness, the rewards obtained by the agent often fluctuate during the initial stages of training.

In the given scenario, the RL agent experiences **fluctuating rewards during the early episodes**. This behavior is expected because the agent is still exploring the environment, collecting information, and learning which actions produce better results. During this stage, the agent has not yet developed a reliable policy, so its performance varies from one episode to another.

As training continues, the agent gradually updates its value estimates and improves its policy based on accumulated experience. By **500 episodes**, the rewards become stable. This stabilization indicates that the agent has learned a consistent strategy for selecting actions that produce reliable rewards.

Stable rewards demonstrate that the policy has improved significantly. The agent no longer depends heavily on random exploration and instead consistently chooses actions that maximize cumulative rewards. This shows that the learning process has become more efficient and the agent has successfully adapted to the environment.

Reward stabilization also indicates **convergence**. Once convergence is achieved, additional training results in only small improvements because the agent has already learned an effective policy. Stable performance suggests that the algorithm has minimized unnecessary exploration and is making optimal or near-optimal decisions.

Another important aspect is **robustness**. Even though the environment is stochastic and contains uncertainty, the agent continues to achieve consistent rewards. This demonstrates that the learned policy is robust and capable of handling unpredictable situations without significant performance degradation.

Therefore, the transition from fluctuating rewards to stable rewards after **500 episodes** clearly indicates successful policy improvement, convergence of the learning algorithm, and robustness of the Reinforcement Learning agent. The observed trend confirms that the agent has learned an effective strategy for making reliable decisions in a stochastic environment.